SOMAIYA VIDYAVIHAR UNIVERSITY

05.05.2025 (E)

Semester:– January 2025 – April 2025 Maximum Marks: 50 Examination: ESE Examination		Duration:2 Hrs.
Programme code: 01 Programme: Btech Comp	Class: TY	Semester: VI(SVU 2020)
Name of the Constituent College:/School K. J. Somaiya School of Engineering	Name of the department: Computers	
Course Code: 116U01C601	Name of the Course: Digital Signal and Image Processing	

Instructions: 1)Draw neat diagrams 2) All questions are compulsory
3) Assume suitable data wherever necessary

Question No.	SECTION B	Marks
Q 4.1	Which transform is commonly used in image compression? a) Laplace Transform b) Z-Transform c) Fourier Transform d) D. Discrete Cosine Transform (DCT)	01 M
Q 4.2	A second order derivative operator can be defined as _____. a) Laplacian b) Gaussian c) Histogram d) Gradient	01 M
Q 4.3	Which of the following filter possess lower frequency? a) High pass filter b) Bandpass filter c) Low pass filter d) No pass filter	01 M
Q 4.4	What is the name of the class that accepts an image's separation of luminesce and reflectance component? a) Homomorphic system b) Base class system c) Base separation system d) Laplacian System	01 M
Q 4.5	Which of the following is a spatial domain technique in image processing? a) DCT b) DFT c) Histogram Equalization d) D. Wavelet Transform	01 M
Q 4.6	An alternate approach used in place of the median filtering is: a) Sharpening b) Laplacian filter c) Gaussian filter d) d. Using a mask	01 M

Q 4.7	Which of the following is the abbreviation of JPEG? a) Joint Photographic Experts Group b) Joint Photographs Expansion Group c) Joint Photographic Expanded Group d) Joint Photographic Expansion Group	01 M
Q 4.8	Gamma Correction is defined as _____ a) Light brightness variation b) A Power-law response phenomenon c) Inverted Intensity curve d) Diverted Curve	01 M
Q 4.9	State different types of Point Processing Techniques.	02 M
Q5 (a)	Given the following 5×5 grayscale image, apply a 3×3 median filter to all pixels except the border pixels. $I = \begin{bmatrix} 10 & 200 & 10 & 200 & 10 \\ 200 & 255 & 50 & 255 & 200 \\ 10 & 50 & 0 & 50 & 10 \\ 200 & 255 & 50 & 255 & 200 \\ 10 & 200 & 10 & 200 & 10 \end{bmatrix}$ **OR**	10 M 10 M
Q5 (a)	Compute the discrete cosine transform (DCT) of the following sequence: **f(x) = {1,2,4,7}**	10 M
Q5 (b)	Describe JPEG compression in detail with necessary block diagram	10 M
Q6 (a)	Given four points in xy plane with the following coordinates (1,1) ,(2,2),(3,3),(4,4) use Hough transform to join these points.	10 M
Q6 (b)	Differentiate between lossy and lossless compression techniques(8 points)	10 M
	OR	
Q6 (b)	Solve any two	05 M
(i)	Hit and Miss Transform	05 M
(ii)	Compare and contrast point and neighborhood spatial domain enhancement techniques.	05 M
(iii)	Hoteling Transform	05 M

Q.

Q 4.6